

Solitito — Project Summary

A real-time guitar chord recognition system

Version 0.2.0, August 2026

1. System overview

Solitito is a real-time guitar trainer implemented in Rust. The program takes a signal from a microphone or audio interface, recognises the material being played, and guides the user through jazz standards, intervals, scales and arpeggios.

Recognition is performed by a neural network of 7.3 million parameters exported to the ONNX format. All processing — DSP, inference and the user interface — is carried out locally on the CPU, without a network connection and without external services.

Four modes of operation are provided:

- **Chords** — complete jazz standards. Green indicates an exact match, yellow a triad or a common substitution, red a chord that was recognised but at a signal level too low to confirm.
- **Intervals** — chord tones played individually, with a selectable set of degrees to practise.
- **Scales** — sequential traversal of the notes defined by a scale.
- **Arpeggios** — chord tones in sequence over a given progression.

Work on the project began in December 2025. This document presents the system architecture, the course of the work, and the design decisions together with their justification.

2. Methodology

The project adopted the principle that every change requires justification by measurement rather than by hypothesis. In practice this meant developing a set of **probes** — scripts that answer a single question at low computational cost, without requiring retraining.

probe	question addressed
<code>verify_annotations.py</code>	do the labels describe the audio content?
<code>probe_root.py</code>	how often does the labelled root actually sound within the window?
<code>probe_sources.py</code>	which GuitarSet chord annotation is usable?
<code>probe_quality.py</code>	from which source should chord quality be derived?
<code>inspect_jams.py</code>	what do the JAMS files actually contain?

The methodology proved effective on repeated occasions. Its converse should also be noted: **hypotheses formulated prior to measurement proved to be wrong in a systematic manner**. These cases are catalogued in Chapter 9.

3. The synthetic dataset

The script `dataset_generator_v2.py` produces, in a single pass, a Guitar Pro file **together with** a complete set of annotations.

3.1. Contents

394 blocks of 6 seconds each (3 bars at 120 BPM), comprising:

- 12 roots $\times$ {maj, min, maj7, dom7, min7, m7b5, dim7, sus4, aug} in several positions on the fretboard,
- all 96 single notes (6 strings $\times$ 16 frets).

Block structure: the first bar constitutes the attack, the second the sustain (tie), and the third silence. The annotation covers the interval `[start + 0.05 s, +3.2 s]`, that is the attack together with the sustain, excluding the decay tail.

3.2. Verification of chord shapes

Before generation begins, the script checks **each of the 21 movable shapes at every fret**, verifying that the shape genuinely produces the declared intervals. An error in the shape table halts generation rather than propagating into the dataset.

3.3. Rendering

The guitar track is exported as a DI signal and rendered in a DAW using [NAM](#) in two variants:

- `synth_dataset_clean.wav` — Fender Deluxe Reverb, clean tone,
- `synth_dataset_eob.wav` — at the edge of breakup.

The recommended sample rate is **48 kHz**, for two independent reasons. NAM operates natively at 48 kHz, so at 44.1 kHz the plugin performs internal resampling. Furthermore, decimation to the 16 kHz at which the model operates is exact at 48 kHz ($48000/16000 = 3$) and is not exact at 44.1 kHz (2.75625).

3.4. Calibration and verification

The `--calibrate <wav>` mode determines the position of the first attack, in case the DAW has added silence at the beginning of the file. PCM 16/24/32-bit and floating-point 32/64-bit formats are supported, in mono and stereo, by means of a self-contained WAV reader with no external dependencies.

The script `verify_annotations.py` compares the label against the **actual audio content**: the dominant pitch class within the window is set against the root indicated by the label. Its only dependency is numpy.

Reference values obtained from the v2 render:

	<code>clean</code>	<code>eob</code>	<code>random baseline</code>
--	--------------------	------------------	------------------------------

<code>top1</code>	87%	77%	~8%
-------------------	-----	-----	-----

<code>top3</code>	100%	98%	~25%
-------------------	------	-----	------

The top3 figure is the meaningful one, since within a chord the root is frequently quieter than the third or the fifth. A value of `top3 > 75%` qualifies the dataset for training; `top3 ≈ 25%` indicates that the labels do not describe the signal.

The script additionally verifies the block period derived from the energy envelope, together with the offset between the start of the annotation and the attack. The two tests are **complementary**: an annotation shifted by an integer multiple of the block length coincides with the attack of an adjacent block and remains undetectable by the timing test. Only a comparison of the labels against the signal content will reveal it.

An overriding principle was adopted: **the generator writes labels directly, and an**

independent script verifies their agreement with the audio. No processing step reconstructs labels from the signal.

4. The GuitarSet dataset

GuitarSet comprises 360 recordings with annotations in JAMS format, captured with a hexaphonic pickup. It is the only source of material from a real instrument used in the project. Bringing it into a usable form required four training runs.

Four properties of the dataset are described below, the omission of which reduces model accuracy.

4.1. Half of the dataset contains no chords

Every excerpt was recorded twice: as `_comp` (accompaniment) and as `_solo` (monophonic improvisation). **The chord annotation is identical in both cases** — it describes the progression over which the performer improvised.

Training the chord heads on solo files amounts to teaching the model that a single note constitutes a complete jazz chord. The solo material comprises 180 of the 360 files.

Filtering these files moved the `Exact` figure from **44.8% to 82.3%** in a single run.

The **pitch** targets derived from solo files remain entirely correct — this is genuine monophonic playing with accurate note annotations, and therefore material well suited to a single-note detector. The trainer consequently retains the pitch loss on solo windows and masks the root and quality only (`GUITARSET_SOLO_MODE = "mask_chord"`).

An indirect consequence should be noted. The `probe_root.py` probe initially determined root audibility across all 360 files, yielding a ceiling of 64.1%. On this basis it was concluded that jazz performers employ rootless voicings and that the chord name is not a function of the signal. That conclusion proved incorrect: once the solo material was filtered out, the root sounds in **97%** of accompaniment windows.

4.2. Chord annotations exist in two variants

Every file carries an `instructed` annotation (the chord as written) and a `performed` annotation

(a transcription of the performance). The number of segments is identical; the distribution of labels is not:

quality	instructed	performed	difference
maj	2640	2106	-534
min	960	460	-500
min7	0	360	+360
maj7	0	430	+430
dom7	480	694	+214
m7b5	240	134	-106
sus	0	132	+132
total	4320	4320	0

The segment total is unchanged, and the choice is therefore not one between more and less data, but a **relabelling of the same recordings**.

The `min` and `min7` rows must be read together: **five hundred segments written as `m` were performed as `m7`**. Training on the `instructed` annotation teaches the model to classify a voicing containing a minor seventh as a plain minor chord. This error was subsequently observed in the application, where the chord `Gm7` was recognised as `Gm`.

Furthermore, the `instructed` annotation contains **not a single** instance of the `maj7` and `min7` classes. Until the switch was made, both classes originated exclusively from the two renders of the synthetic dataset, that is from a single instrument processed through a single amplifier. This produced a figure of 100% on the validation set (the same instrument on both sides of the split) together with an absence of robustness on real material.

Switching to the `performed` annotation moved the `Exact` figure from **82.3% to 92.4%**. Root accuracy was unaffected: the two annotations differ with respect to the root in **0 out of 43,056** comparisons.

4.3. The split must follow the source

Randomly shuffling the list of chord segments and splitting it 94/6 places adjacent bars of **the**

same recording on both sides of the split — with the same instrument, room, microphone and take, and frequently with the same chord recurring one bar later.

In the synthetic dataset the dependency is stronger still: the `clean` and `eob` renders of one block constitute the same performance processed through a different amplifier, and were assigned to the training and validation sets independently.

The solution adopted is grouping by source: the entire file for GuitarSet and the entire block (both renders) for the synthetic dataset.

All validation figures fall following this change. This does not constitute a regression of the model, but the removal of an inflation that invalidated earlier conclusions regarding generalisation. The value `root_acc = 98%` obtained in run `take1` was to a large extent an artefact: with `both` annotations the same segment occurred twice, so an identical window entered both the training and the validation set.

4.4. Pitch targets derived from `note_midi`

The chord annotation describes the **intended harmony** over a span of several seconds. The training window covers 0.77 s and frequently does not contain the labelled seventh at all. The model was therefore penalised for failing to predict a note absent from the signal — seventh recall on GuitarSet stood at 32%.

The hexaphonic pickup provides `note_midi` annotations describing the actual performance on each string separately. Deriving the pitch targets from these raised seventh recall from **32% to 96%**.

The threshold adopted: a note must sound for at least 25% of the window (`NOTE_MIN_COVER`) in order to be included in the target.

5. Architecture

5.1. Signal path

```
audio input → resampling to 16 kHz → FFT (8192) → sparse pseudo-CQT → features → ONNX model
```

Resampling to 16 kHz. The CQT spans 6 octaves from C1, so the highest bin falls in the region

of 2 kHz, well below the 8 kHz Nyquist limit. Bandwidth does not constitute a constraint.

Pseudo-CQT. In place of a true constant-Q transform, the application multiplies the FFT spectrum by a precomputed kernel: 144 bins, 24 per octave, corresponding to quarter-tone resolution. The kernel is obtained from `librosa.filters.constant_q`, by which means the application and the trainer produce identical features.

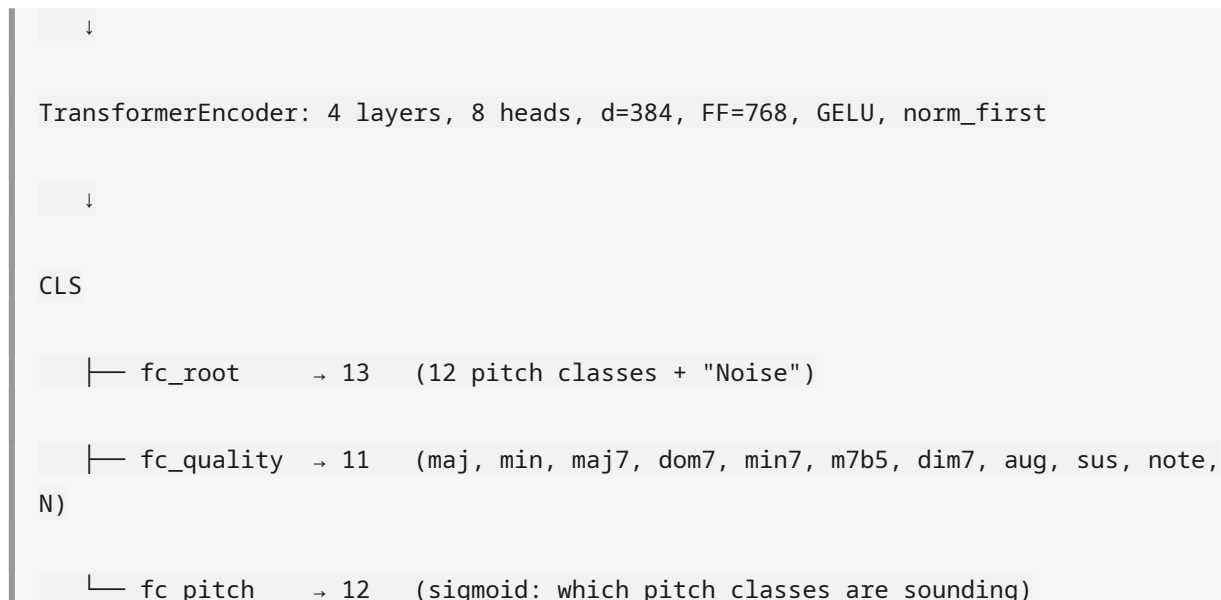
Features. 168 values per frame:

range	contents
0–143	CQT bins after log normalisation
144–155	chroma (<code>cq_to_chroma</code> matrix, per-frame maximum normalisation)
156–167	bass energy (mean of bin pairs 0–23)

The model spans 48 frames of history at a hop of 256 samples, corresponding to **0.77 s**.

5.2. Network

```
input [48, 168]
↓
InstanceNorm2d
↓
ConvBlockSE 1 → 48 (Squeeze-and-Excitation)
ConvBlockSE 48 → 96
ConvBlockSE 96 → 192
ConvBlockSE 192 → 384
↓
Linear 3840 → 384
↓
+ CLS token, positional encoding
```



Total parameter count: 7,286,038.

5.3. Division of tasks between the heads

The distinction between the roles of the individual heads is of central importance and has been confirmed by measurement.

head	result	role
pitch_logits	F1 0.909	which notes are sounding — the basis of the Intervals, Scales and Arpeggios modes
root_logits	98.1%	the name of the root
quality_logits	~93%	the chord family

An early version of the application derived chord quality from the pitch vector using manually determined thresholds. The `probe_quality.py` probe compared three methods on the same checkpoint:

method	accuracy
the <code>quality_logits</code> head	80.5%
template matching against the predicted pitch	66.0%
template matching against the true pitch vector	59.2%

The head exceeds template matching against a *precisely known* set of notes by 21 percentage

points. It therefore extracts from the signal information not present in the set of pitch classes itself: timbre, the distribution of the voicing across the register, and the shape of the attack.

Design conclusion: the quality head remains a necessary component.

6. Training

6.1. Phases

The training procedure comprises three phases, of which the first is of principal significance.

phase	scope	status
1	main training, 120 epochs, cosine LR with warm-up	the only phase yielding improvement
2	tuning of the pitch head threshold	corrected — it had sorted by a metric independent of the threshold
3	head fine-tuning with the encoder frozen	disabled

Phase 2 scanned thresholds over the range 0.30–0.70, optimising the `exact` figure. That figure is the conjunction of `argmax(root)` and `argmax(quality)`, and therefore took an identical value across all 41 thresholds, making the selection arbitrary. Sorting is now performed by the F1 of the pitch head, on which the threshold genuinely bears.

Phase 3 was disabled after its effects were measured over three consecutive runs:

```
take2, 40 epochs: pitch_f1 0.9318 → 0.9326 (+0.0008), exact 0.5455 → 0.5445

take3,  4 epochs: F1 0.933 → 0.931,          exact 54.6% unchanged
```

The encoder remains frozen and only the heads are trained, at a learning rate of 1e-5. The phase possesses no mechanism by which to improve the model, and its cost amounts to approximately 1.5 hours of computation.

6.2. Loss functions and masking

- **root** — CrossEntropy with label smoothing of 0.05,
- **quality** — CrossEntropy with smoothing, sampler weighted by class,
- **pitch** — Focal BCE ($\gamma = 2.0$, `pos_weight` 2.5), auxiliary weight 0.7.

Two masking mechanisms were applied, both justified by measurement.

`MASK_ROOT_WHEN_SILENT` — the root loss is computed exclusively on windows in which the root genuinely sounds. Training the root on windows in which it is absent does not result in learned perception, but in memorisation of the GuitarSet progressions, while the shared encoder receives a gradient in conflict with the pitch target.

`GUITARSET_SOLO_MODE = "mask_chord"` — the root and quality receive no gradient from solo recordings; the pitch head receives it unchanged.

6.3. Augmentation

- **pitch shift** by $\pm N$ semitones. An implementation detail of consequence: the CQT and the bass energy are shifted **with zero fill**, whereas the chroma is shifted **cyclically**. The chroma is circular by definition; the CQT is not, and wrapping the bass band around to the top of the range would introduce notes absent from the signal.
- **time and frequency masking** (SpecAugment),
- **spectral tilt and noise** — simulation of varied signal chains.

6.4. Energy gate

The parameter `ENERGY_KEEP_FRAC = 0.55` rejects windows whose energy falls below 55% of the segment peak. The justification: during the decay phase the seventh, being the quietest component of the voicing, is the first to disappear, while the label remains unchanged. In the absence of the gate this would systematically train the collapse `m7 → m`.

6.5. Metrics

Chord metrics are computed **exclusively on windows in which the label describes the signal**, with solo windows excluded. They are additionally reported separately for windows with and without an audible root, since the combined figure conflates two distinct populations.

Selection of the best checkpoint proceeds by the figure `composite = (root_audible + qual +`

exact) / 3. Use of the combined root_acc would favour a model that reproduces progressions effectively over one that analyses the signal correctly.

The TRAIN diagnostic check, performed every 5 epochs, computes metrics on the training data without augmentation. It addresses the question of whether the model is capable of reproducing its own training data. A negative answer identifies the features or the labels as the source of the constraint rather than generalisation, and indicates that increasing the number of epochs would serve no purpose.

7. Results

Model v2_take6, validated with a split by source and with solo windows excluded:

metric	value
root accuracy	98.1%
pitch F1	0.909
exact match (root and quality)	92.4%

Accuracy by quality at the best checkpoint: dom7 97%, min7 93%, min 92%, sus 91%, maj 89%, maj7 89%; the classes m7b5, dim7 and aug above 97%.

7.1. Course of the work

run	change	Exact
take1–take3	various, on a split subject to leakage	not comparable
take4	split by source — reference point	44.8%
take5	masking of solo recordings	82.3%
take6	performed annotations	92.4%

7.2. Limiting factor

The difference between the training and validation sets with respect to quality amounts to 6.5 percentage points (99.2% against 92.7%). The model reproduces the training data. This

corresponds to the profile of a constraint imposed by **generalisation** rather than by the capacity of the architecture.

Practical conclusion: neither an increase in the number of epochs nor an increase in model size will yield improvement. Improvement will follow from an increase in the quantity of varied material from a real instrument.

8. The application

Recognition accuracy is not equivalent to the usefulness of a trainer. Three matters proved to carry weight comparable to changes in the model.

8.1. Requirement for a full context window

The trainer determined windows **exclusively within** the sustaining chord (`range(start, end - 48)`). Following a strike of the strings, the application buffer contains partial silence for 0.77 s; taking the FFT window into account (8192 samples, that is 512 ms), the oldest frame describes signal from as much as 1.3 s earlier. This constitutes an input outside the training distribution.

The observed symptom: seventh chords were recognised only during the sustain phase, that is at the first moment at which the window becomes entirely filled with the chord.

The solution adopted: the application issues no query to the model until the window is 90% filled with signal.

8.2. Differing decay rates of chord components

Diagnostic output for a sustained `Gm7` chord:

```
G m7 | min7=96% | b7=96      ← immediately after the strike

G m7 | min7=82% | b7=76

G m7 | min7=52% | b7=52

G m  | min=49%  | b7=45      ← the seventh has faded, the model reclassifies
```

The model's classification is correct — the seventh is genuinely no longer present in the

current window. A chord does not, however, change its identity in the course of sustaining.

The solution adopted: a quality latch. The mechanism **engages** at a confidence of ≥ 0.60 but **holds its state independently of confidence**. During the decay phase the model reports the reduced quality at a confidence in the region of 94–96%, so a confidence threshold alone would afford no protection. The latch is released by a new attack or by a change of root.

The latch engages only 48 frames after the attack. Before that point the window still contains the tail of the **preceding** chord, which would result in an incorrect name being latched.

Attack detection compares the signal level against a slowly moving envelope (EMA) rather than against an absolute threshold, which would depend on playing volume. A refractory period of 0.2 s was additionally applied so that a single strum triggers exactly one attack.

8.3. Measuring time rather than assuming it

The progress counter received the fixed value `dt = 0.040`, whereas the inference thread required 55–90 ms per cycle (inference together with 40 ms of sleep). The counter therefore ran slower than real time: a threshold of 0.6 s was reached after approximately one second, with the value depending on machine load.

Following correction — comprising measurement of real elapsed time, determination of the period by the inference thread from the start of the cycle, narrowing of the voting window from 5 to 3, and reduction of the default threshold to 0.25 s — the transition takes **approximately 0.3 s** in place of 1.2 s.

8.4. Sparse representation of the CQT kernel

The full kernel comprises $4097 \times 144 = 589,968$ weights, concentrated around the centre frequency of each bin. Discarding weights below $1e-4$ of the peak value yields the following results:

threshold	weights retained	max. error relative to peak
1e-5	21.9%	0.006%
1e-4	6.9%	0.033%
1e-3	2.3%	0.352%

The error was determined on three spectra: white, pink, and a guitar-like harmonic series.

The CQT is followed by log normalisation over an 80 dB range, so a figure of 0.03% remains orders of magnitude below the resolution of the feature.

The size of the weights file falls from **28 MB to 2 MB**, and the audio path performs approximately **14 times fewer multiplications** per frame. The previous implementation traversed all 4097 FFT bins for each of the 144 CQT bins, discarding zero values only within the loop.

8.5. Feature agreement between the trainer and the application

Discrepancies of this class are particularly difficult to diagnose, since the application remains functional and exhibits only classification errors. Two were identified:

- **chroma mapping.** The file distributed with the application folded bins in the pairs $(0,1), (2,3), \dots$, whereas `librosa.cq_to_chroma` applies the division $(1,2), (3,4), \dots$. Every second bin was assigned to the adjacent class, which corresponds to a chroma blurred by a semitone across half the band.
- **cache key.** The cache file name was derived from the expression `abs(hash(path))`. The Python interpreter randomises the hash seed for strings on every process start, so the cache was never reused between sessions. An SHA-1 digest of the file name is now employed.

The application **rejects** weights in the previous, dense format, reporting this by means of a message.

8.6. User interface

Following a review, the number of controls was reduced from five to four:

control	remarks
Noise gate	in dBFS, with a level meter on the same scale and a threshold marker
Chord confidence	threshold for the chord name (Chords mode)
Note threshold	threshold for a single note (note-based modes)
Hold time	the period for which a correct chord must be sustained

The `Tail` control (set from the interface and read nowhere in the code) and the `In gain` control (whose effect was cancelled by the per-frame normalisation, so that it shifted only the

same inequality as the noise gate) were removed.

The `Confidence` control governed two distinct quantities simultaneously, and in the note-based modes was subject to a lower bound of `.max(0.5)`, as a result of which the entire range 0.1–0.5 produced identical behaviour. The functions were separated.

The noise gate previously operated on a linear RMS scale of 0–0.1, which **did not reach the noise level of a laptop microphone** (RMS 0.05–0.15 after gain). A decibel scale of –72...0 dBFS provides resolution across the required range together with coverage to full scale.

8.7. Diagnostic mode

```
| SOLITITO_DEBUG=1 ./solitito
```

The mode prints, on each prediction, the three strongest qualities together with the pitch vector expressed as **intervals relative to the recognised root**:

```
| G m7 | min7=97% sus=0% maj=0% | R96# b25 28 b382# 37 44 b56 594# b616 69 b797# 74
```

The tool distinguishes the case in which the model fails to detect the seventh from the case in which it detects it but disregards it in classification. Both symptoms are indistinguishable at the level of the chord name and call for opposite corrective actions. The `Gm7` case was resolved by these means without retraining.

9. Hypotheses refuted by measurement

This chapter documents cases in which measurement refuted a previously held assumption.

hypothesis	measurement result
A normalisation discrepancy (per frame against global) constitutes the principal constraint	difference below 1 pp; the <code>InstanceNorm2d</code> layer compensates for it
A harmonic barrier — the third harmonic of b3 coincides with b7, rendering min7 and min indistinguishable	the measurement had been performed on segments carrying random labels

The model does not reproduce the training TRAIN check: 83.7% against 63.1% on data (underfitting) validation, hence overfitting

Template matching will outperform the quality head (B $\approx$ 75% against A = 63%) B = 61.0%

The 64% root ceiling follows from rootless jazz an artefact of the solo recordings; 97% in voicings accompaniment

Masking the root will unblock quality beyond 73% quality remained at 72%

The chroma in the distributed file is one-hot `cq_to_chroma` at 24 bins per octave likewise assigns one weight per bin; the discrepancy and therefore defective concerned a shift

Without the `ORT_DYLIB_PATH` variable the `RUNPATH=$ORIGIN` from the `.cargo/config.toml` binary will use the system library file already addressed this

The pattern is unambiguous: **measurement results held consistently, whereas predictions formulated prior to measurement proved wrong in a systematic manner.** This justifies the adopted methodology based on probes.

10. Design decisions

10.1. Solutions adopted

The generator writes labels directly. No processing step reconstructs information from the signal.

Verification of labels by an independent script. A measure redundant with respect to the generator, applied deliberately.

Splitting the dataset by source. It lowers the reported figures by more than ten percentage points and is justified.

Three heads with separated roles. The note-based modes rely on the pitch vector rather than on the chord name.

A sparse representation of the CQT kernel. The benefit is twofold: the size of the weights file and the processing time in the audio thread.

Rejection of incompatible weights by the application. Silent acceptance would result in a program that is functional but misclassifies.

Compiled-in strings, without the gettext library. With several dozen strings, a system dependency and `.mo` catalogues incur a cost exceeding the benefit.

10.2. Solutions rejected

Deriving quality from the pitch vector — measured as 21 percentage points inferior to the quality head.

Temporal aggregation as a means of improving quality — on a controlled population it yields approximately +1 pp. The model's errors are correlated in time: the model exhibits no vacillation between windows, but indicates the same incorrect answer consistently and with high confidence.

A bar-based mode (a score advancing in tempo, with assessment in place of a gate) — implemented and subsequently withdrawn. It was determined that the trainer is to respond dynamically.

Training phase 3 — measured as yielding no improvement across three runs.

10.3. Open matters

- **A test set from the target instrument.** All figures relate to six external performers and two renders of the synthetic dataset. No measurement on the target signal chain is available.
 - **Changing `CTX_FRAMES` from 48 to 32** — the only architectural modification with genuine justification (latency), to be verified in a single run.
 - **Increasing the quantity of material from a real instrument** — the only factor capable of reducing the 6.5 percentage point difference.
-

11. Conclusion

The work carried out resulted in a model achieving 92.4% exact matches on validation determined with a split by source, released as distribution packages for two platforms.

The principal gain in accuracy followed not from changes to the architecture, but from four findings concerning the data:

1. half of the GuitarSet dataset consists of improvisation labelled with accompaniment chords,
2. the `instructed` annotation contains no sevenths and misclassifies five hundred segments,
3. splitting the dataset at the segment level introduces leakage,
4. pitch targets must be derived from the actual performance rather than from the score.

These four changes moved the `Exact` figure from 44.8% to 92.4%. None of them concerned the structure of the network.

This document describes the state as of August 2026, version 0.2.0. Repository: <https://github.com/greblus/solitito>